

## HANCOCK COUNTY BAR HARBOR AP, ME, USA

WMO: 726077

Lat: 44.450N

Lon: 68.367W

Elev: 88

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14616

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -2.0       | 1.9 | -17.5                         | 2.1 | 1.2  | -11.0 | 3.0 | 5.2  | 32.2                  | 35.6 | 26.3 | 41.0 | 6.0                   | 320  | 0.580 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.1                   | 81.7            | 68.7 | 79.1 | 66.7 | 75.4 | 65.0 | 71.5                | 78.4 | 69.7 | 75.5 | 67.8 | 73.1 | 8.8                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.8                            | 110.0 | 75.7 | 67.6 | 101.7 | 72.8 | 65.8 | 95.6 | 71.2 | 35.3          | 78.6 | 33.7 | 75.6 | 32.3 | 73.4 | 78.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c)  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 22.0              | 18.7 | 16.6 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |      | -7.2                       | 88.1 | 5.6                | 3.3 | -11.2                                              | 90.4 | -14.5      | 92.3 | -17.6      | 94.1 | -21.6      | 96.5 |  |  |
|                   |      |      | -8.3                       | 74.6 | 5.0                | 1.7 | -11.9                                              | 75.8 | -14.8      | 76.8 | -17.6      | 77.8 | -21.3      | 79.0 |  |  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 44.9   | 21.7  | 24.3  | 31.6  | 41.8  | 51.5  | 59.9  | 66.5  | 65.8  | 59.1  | 48.6  | 38.4  | 28.8 |
|                                                                               | DBStd       | 17.08  | 11.05 | 9.89  | 8.34  | 6.01  | 5.65  | 5.54  | 4.40  | 4.31  | 5.87  | 6.77  | 8.07  | 9.44 |
|                                                                               | HDD50       | 3598   | 878   | 720   | 572   | 253   | 46    | 2     | 0     | 0     | 4     | 108   | 357   | 658  |
|                                                                               | HDD65       | 7502   | 1343  | 1140  | 1035  | 696   | 421   | 168   | 34    | 41    | 193   | 509   | 799   | 1123 |
|                                                                               | CDD50       | 1752   | 0     | 0     | 1     | 7     | 92    | 300   | 513   | 490   | 276   | 65    | 8     | 0    |
|                                                                               | CDD65       | 180    | 0     | 0     | 0     | 0     | 2     | 16    | 81    | 66    | 15    | 0     | 0     | 0    |
|                                                                               | CDH74       | 1076   | 0     | 0     | 0     | 0     | 35    | 129   | 471   | 353   | 83    | 5     | 0     | 0    |
|                                                                               | CDH80       | 183    | 0     | 0     | 0     | 0     | 6     | 23    | 82    | 58    | 14    | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 7.1    | 7.4   | 7.5   | 8.2   | 8.2   | 7.4   | 6.6   | 6.1   | 5.8   | 6.3   | 7.2   | 7.2   | 7.0  |
| Precipitation                                                                 | PrecAvg     | 54.1   | 4.4   | 3.7   | 4.8   | 4.7   | 4.2   | 4.1   | 3.2   | 2.8   | 4.4   | 5.7   | 5.8   | 5.6  |
|                                                                               | PrecMax     | 75.1   | 9.8   | 8.3   | 8.4   | 9.6   | 11.9  | 11.0  | 7.1   | 7.1   | 10.3  | 15.4  | 11.3  | 10.4 |
|                                                                               | PrecMin     | 28.6   | 0.4   | 1.3   | 1.1   | 0.3   | 0.9   | 1.6   | 1.0   | 0.6   | 1.1   | 1.3   | 2.7   | 2.2  |
|                                                                               | PrecStd     | 9.2    | 2.1   | 1.6   | 2.2   | 2.4   | 2.5   | 2.4   | 1.7   | 1.8   | 2.4   | 3.0   | 1.9   | 1.9  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 51.6  | 50.6  | 57.5  | 65.9  | 79.4  | 83.8  | 86.4  | 85.8  | 81.3  | 72.0  | 61.3  | 54.4 |
|                                                                               |             | MCWB   | 50.4  | 47.4  | 47.2  | 51.3  | 62.5  | 70.6  | 71.6  | 70.2  | 68.4  | 60.1  | 56.6  | 51.9 |
|                                                                               | 2%          | DB     | 46.1  | 45.3  | 51.3  | 59.1  | 71.6  | 78.5  | 81.8  | 81.1  | 75.3  | 65.3  | 57.3  | 50.4 |
|                                                                               |             | MCWB   | 43.9  | 41.5  | 43.3  | 47.3  | 58.9  | 64.4  | 69.0  | 68.9  | 65.3  | 58.9  | 53.9  | 48.6 |
|                                                                               | 5%          | DB     | 42.8  | 41.3  | 46.0  | 55.0  | 65.8  | 73.4  | 79.2  | 78.5  | 72.2  | 62.7  | 54.5  | 46.3 |
|                                                                               |             | MCWB   | 39.9  | 37.9  | 40.9  | 46.1  | 55.2  | 61.6  | 67.1  | 66.7  | 63.3  | 57.6  | 51.1  | 44.0 |
|                                                                               | 10%         | DB     | 37.3  | 38.5  | 43.2  | 52.3  | 62.5  | 71.5  | 75.5  | 74.9  | 69.6  | 59.4  | 52.0  | 43.4 |
|                                                                               |             | MCWB   | 34.7  | 35.3  | 38.0  | 44.9  | 53.0  | 61.3  | 65.8  | 65.1  | 62.4  | 54.6  | 48.8  | 40.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 49.7  | 47.4  | 50.6  | 55.3  | 65.9  | 71.9  | 74.1  | 74.2  | 72.0  | 64.7  | 58.4  | 53.1 |
|                                                                               |             | MCDB   | 50.9  | 48.2  | 55.8  | 60.4  | 75.1  | 81.2  | 82.3  | 81.6  | 78.7  | 65.8  | 60.5  | 53.7 |
|                                                                               | 2%          | WB     | 45.2  | 42.7  | 45.9  | 51.6  | 60.7  | 67.6  | 71.5  | 71.4  | 68.6  | 61.6  | 55.1  | 48.6 |
|                                                                               |             | MCDB   | 46.0  | 44.8  | 47.7  | 54.4  | 67.4  | 73.8  | 78.5  | 77.6  | 72.6  | 64.0  | 57.0  | 49.8 |
|                                                                               | 5%          | WB     | 40.0  | 38.6  | 42.1  | 48.4  | 57.2  | 65.0  | 69.8  | 69.5  | 65.9  | 58.6  | 52.1  | 45.5 |
|                                                                               |             | MCDB   | 42.4  | 40.9  | 45.6  | 52.9  | 63.7  | 70.7  | 76.0  | 74.6  | 69.5  | 61.9  | 54.0  | 46.2 |
|                                                                               | 10%         | WB     | 35.5  | 35.9  | 38.8  | 46.1  | 54.5  | 62.9  | 67.9  | 67.5  | 63.7  | 55.5  | 48.3  | 40.8 |
|                                                                               |             | MCDB   | 37.5  | 38.1  | 42.5  | 50.2  | 59.3  | 67.3  | 73.6  | 72.3  | 66.2  | 58.7  | 50.8  | 43.6 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 16.1  | 16.4  | 15.0  | 15.8  | 16.7  | 16.7  | 17.1  | 17.5  | 16.9  | 15.6  | 13.9  | 14.7 |
|                                                                               |             | MCDBR  | 20.0  | 16.6  | 18.5  | 21.8  | 24.8  | 22.1  | 21.3  | 21.0  | 20.5  | 18.1  | 16.4  | 18.6 |
|                                                                               |             | MCWBR  | 20.1  | 14.4  | 13.6  | 13.4  | 13.6  | 11.8  | 10.7  | 11.0  | 12.3  | 12.9  | 14.8  | 18.2 |
|                                                                               | 5% WB       | MCDBR  | 20.2  | 16.2  | 16.7  | 17.6  | 20.4  | 18.3  | 17.8  | 17.1  | 16.8  | 16.0  | 14.8  | 18.8 |
|                                                                               |             | MCWBR  | 20.8  | 15.4  | 14.3  | 13.0  | 13.3  | 11.9  | 10.6  | 10.5  | 11.8  | 13.6  | 15.2  | 19.2 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.289  | 0.300 | 0.323 | 0.352 | 0.376 | 0.416 | 0.430 | 0.404 | 0.362 | 0.332 | 0.314 | 0.295 |      |
|                                                                               | taud        | 2.480  | 2.465 | 2.474 | 2.429 | 2.402 | 2.307 | 2.286 | 2.348 | 2.448 | 2.531 | 2.547 | 2.509 |      |
|                                                                               | Ebn at Noon | 267    | 284   | 291   | 289   | 284   | 272   | 266   | 269   | 273   | 269   | 256   | 253   |      |
|                                                                               | Edh at Noon | 22     | 27    | 30    | 34    | 36    | 40    | 41    | 37    | 31    | 25    | 21    | 20    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 508    | 782   | 1157  | 1437  | 1643  | 1783  | 1855  | 1655  | 1338  | 856   | 526   | 395   |      |
|                                                                               | RadStd      | 46     | 64    | 70    | 150   | 198   | 147   | 137   | 96    | 81    | 80    | 47    | 40    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (26 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page